

Job Description: Machine Learning Engineer / Data Analyst (Analytics and LLMs)

Position Title: Machine Learning Engineer / Data Analyst

Job Location: 5 days working

Job Overview

We are seeking a motivated and detail-oriented Machine Learning Engineer / Data Analyst to build intelligent systems across customer analytics, recommendation engines, and LLM-powered workflows. This role sits at the intersection of data, product, and AI — driving personalization, automation, and smarter decision-making across our platform.

Key Roles and Responsibilities

1. Customer Analytics & Segmentation

- Build advanced customer segmentation models (clustering, behavioral cohorts, customer persona)
- Develop predictive models:
 - Churn prediction
 - Customer lifetime value (LTV)
 - Campaign uplift / response prediction
- Design customer journeys and funnel analytics using large-scale data

2. NLP & LLM Applications

- Build and deploy NLP/LLM-driven solutions:
 - Campaign/content generation
 - Customer communication optimization
 - Chatbots / conversational workflows
- Design and optimize **prompt engineering strategies**
- Fine-tune or adapt LLMs for domain-specific use cases

3. Data & ML Engineering

- Work with large-scale datasets using SQL (ClickHouse preferred)
- Build end-to-end pipelines:
 - Data extraction → feature engineering → model training → deployment
- Integrate ML outputs into BI tools (Power BI, Superset, etc.)
- Ensure scalability and performance of ML systems

Required Experience

- 1 - 3 years of experience

Required Skills

- Strong Python skills (pandas, numpy, scikit-learn)
- Strong SQL (handling large datasets efficiently)
- Solid understanding of:
 - Supervised & unsupervised learning
 - Clustering techniques (K-means, hierarchical, DBSCAN)
 - Recommendation systems fundamentals
- Hands-on experience with **NLP basics**
- Experience with **LLMs and prompt engineering**
- Strong problem-solving and analytical thinking

Preferred Skills (High Impact)

- Experience with **LLM frameworks** (LangChain, LlamaIndex, etc.)
- Familiarity with **vector databases** (FAISS, Pinecone, etc.)
- Experience with **ClickHouse or OLAP systems**
- Knowledge of **MLOps** (Docker, Airflow, MLflow) (optional - if fresher)
- Experience in **CRM / Loyalty / Retail analytics domain** (optional - if fresher)
- Basic frontend/API exposure for ML integration (optional - if fresher)

About the Company:

OptCulture is at the forefront of helping brands elevate their customer relationships through cutting-edge retention tools. We don't just connect the dots; we create journeys that keep customers coming back for more!

Think about the brands you admire - IKEA, Marks & Spencer, GUESS, Style Union, we're the behind-the-scenes marketing technology enabling them to understand and engage with their customers in deeply meaningful ways. It's not just about sales; it's about fostering loyalty that lasts.

What Makes Us Unique?

OptCulture isn't your typical tech company. We're a bootstrapped powerhouse, driven by relentless innovation and determination. **From Houston to Dubai to Hyderabad**, our solutions are redefining customer retention on a global scale. And here's the kicker—we're growing!

OptCulture aims to hire thinkers and achievers. We believe in providing an environment for fast-paced growth as an individual, team, and organization. We encourage a culture of independence, collaboration, trust, and balance.